



```
1  module mealy_Nooverlap(  
2      input clk, reset, x,  
3      output reg z  
4  );  
5      reg [1:0] state, next_state;  
6      parameter S0 = 2'b00, S1 = 2'b01, S2 = 2'b10; S3=2'b11;  
7  
8      always @(posedge clk or posedge reset)  
9          if (reset) state <= S0;  
10         else state <= next_state;  
11  
12         always @(state or x) begin  
13             case (state)  
14                 S0: next_state = x ? S1 : S0;  
15                 S1: next_state = x ? S1 : S2;  
16                 S2: next_state = x ? S3 : S0;  
17                 S3: begin  
18                     next_state = x ? S1 : S0;  
19                     z = x ? 1'b1 : 1'b0;  
20                 end  
21                 default: next_state = S0;  
22             endcase  
23             if (state != S3) z = 1'b0;  
24         end  
25     endmodule
```



```
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2      input clk, reset, x,  
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4  );  
5      reg [1:0] state, next_state;  
6      parameter S0 = 2'b00, S1 = 2'b01, S2 = 2'b10;  
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8      always @(posedge clk or posedge reset)  
9          if (reset) state <= S0;  
10         else state <= next_state;  
11  
12         always @(state or x) begin  
13             case (state)  
14                 S0: next_state = x ? S1 : S0;  
15                 S1: next_state = x ? S1 : S2;  
16                 S2: begin  
17                     next_state = x ? S1 : S0;  
18                     z = x ? 1'b1 : 1'b0;  
19                 end  
20                 default: next_state = S0;  
21             endcase  
22             if (state != S2) z = 1'b0;  
23         end  
24     endmodule
```



```
1  module moore_NOoverlap(  
2      input clk, reset, x,  
3      output reg z  
4  );  
5      reg [1:0] state, next_state;  
6      parameter S0 = 2'b00, S1 = 2'b01, S2 = 2'b10, S3 = 2'b11;  
7  
8      always @(posedge clk or posedge reset)  
9          if (reset) state <= S0;  
10         else state <= next_state;  
11  
12         always @(state or x) begin  
13             case (state)  
14                 S0: next_state = x ? S1 : S0;  
15                 S1: next_state = x ? S1 : S2;  
16                 S2: next_state = x ? S3 : S0;  
17                 S3: next_state = x ? S1 : S0;  
18                 default: next_state = S0;  
19             endcase  
20         end  
21  
22         always @(state) z = (state == S3);  
23     endmodule
```



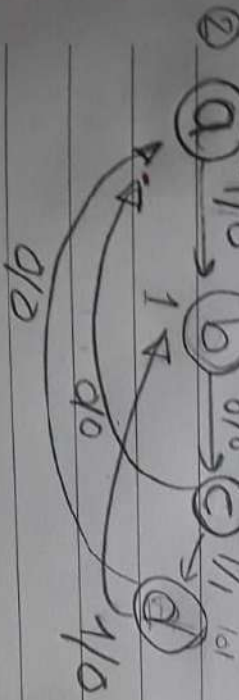
```
1  module moore_overlap(  
2      input clk, reset, x,  
3      output reg z  
4  );  
5      reg [1:0] state, next_state;  
6      parameter S0 = 2'b00, S1 = 2'b01, S2 = 2'b10, S3 = 2'b11;  
7  
8      always @(posedge clk or posedge reset)  
9          if (reset) state <= S0;  
10         else state <= next_state;  
11  
12         always @(state or x) begin  
13             case (state)  
14                 S0: next_state = x ? S1 : S0;  
15                 S1: next_state = x ? S1 : S2;  
16                 S2: next_state = x ? S3 : S0;  
17                 S3: next_state = x ? S1 : S2;  
18                 default: next_state = S0;  
19             endcase  
20         end  
21  
22         always @(state) z = (state == S3);  
23     endmodule
```


Case ① Mealy overlap

Memor No. 101
Date 1 1



Mealy without overlap



③ More without overlap

